

WIFI SECURITY CHECKER PROJECT SUMMARY

This project is a **Python-based Wi-Fi Security Checker** developed to analyze the security status of the currently connected Wi-Fi network. The tool automatically detects the operating system and retrieves Wi-Fi details such as **SSID, signal strength, and encryption type**, then evaluates the security level of the connection.

Initially, the project started as a basic Linux-only Wi-Fi checker that relied on Linux networking commands such as:

- **iwgetid**
- **iwconfig**
- **nmcli**

The original version had several limitations:

- Worked only on Linux systems
- Failed on Windows because Linux commands were unavailable
- iwgetid returned exit status 255 when Wi-Fi was disconnected
- Used deprecated commands like iwconfig
- Used unsafe shell=True execution
- Incorrect WPA detection logic
- Did not detect open networks properly
- Poor exception handling
- Minimal terminal presentation

During testing, the script failed with errors such as:

Error: Command 'iwgetid -r' returned non-zero exit status 255

and on Windows:

[ERROR] nmcli is not installed.

After analyzing and fixing these issues, the project was enhanced into a **fully functional cross-platform Wi-Fi security utility**.

```
(root@kali)-[/home/./Desktop/VIEH_Group/cybersecurity-automation/Day-02-Wifi-Security-Checker]
# python wifi_checker.py

Wi-Fi Security Checker

Error: Command 'iwgetid -r' returned non-zero exit status 255.
```

The final tool can:

- Detect connected Wi-Fi network
- Show signal strength
- Identify encryption type
- Detect WPA / WPA2 / WPA3 / Open networks
- Work on both Windows and Linux

```
C:\Users\zalam\Downloads\Day 2 - Wifi Security Checker>py wifi_checker.py

WIFI

WIFI Security Checker

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Operating System : Windows
Connected Wi-Fi : Airtel_Alam House
Signal Strength : 78%
Encryption Type : WPA2-Personal

[ SECURITY STATUS ]
[+] WPA2 detected (Good Security)

=====
```

```
C:\Users\zalam\Downloads\Day 2 - Wifi Security Checker>py wifi_checker.py

WIFI

WIFI Security Checker

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Operating System : Windows
Connected Wi-Fi : Zeeshan
Signal Strength : 84%
Encryption Type : WPA3-Personal (H2E)

[ SECURITY STATUS ]
[+] WPA3 detected (Strong Security)

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```